bitwize operators

types of bitwise operators AND operators (&) so it will check each bit value is 1 or not. if both are 1 then only i will give output 1

101 5

001 1

---

001 1

or operator it will check any value is 1 or not ,

xor operator ^ if both value are matching it will give you output zero , not matching it will give you 1 as a output

101 5

001 1

---

100 4

complement operator ~5 ==> -(x+1)

x << n ==> x *2*n   left shift operator

x>>n ==> x/2*n right shift operator

complement ~x ==> -(x+1) complement operator

in left shift it will shift all the value to the left ,based on the given frequency number.

so right shift operator will give the value to right ,based on the given frequency number.

ex: 100>>4

now 100 bin code will be shifted to right side 4 times

1100100

after one shift it will be 110010 (100>>1)

2 shift it will bee 11001   (100>>2)

3 shift will be 1100 (100>>3)

4. shift the value will bee

5&4

4

101   AND multiply the number

100

100

5|4

bin 5 == 101 OR multiply the number

bin 4 == 100

101

5^4

1

bin 5 = 101 same value will work as zero

bin 4 = 100

001

~4

left shift

right shift operator

x << n ==> x *2*n

x>>n ==> x/2*n

complement ~x ==> -(x+1)

special operator

identity operator ::: this operator are useful to compare the memory address of the data types

2. a single element data type will have same data address even after assigning it to the different variables

3. multiple element data type will have different address for different variables

single element data type multi element data type

a = [1,2,3]

b = [1,2,3]

print(id(a))

```
print(id(b))
print(a is  b)
print(a == b)   ## identity operator
```

output was

```
1992713176256
1992713251392
True
True
```

single element data type multi element data type

```
a = [1,2,3]
b = [1,2,3]
print(id(a))
print(id(b))
print(a is  b)
print(a == b)   ## identity operator
```

output was

```
1992713176256
1992713251392
```

True

True a = [1,2,3]

b = [1,2,3]

print(id(a))

print(id(b))

print(a is not b)

print(a == b) ## identity operator

output was

1992713176256

1992713251392

False

True

membership operator

in ,not in ::: it will check the value in the inerrable are not

ex: 4 in [1,2,3]

output will be false

things we have learn

1. bitwise operator
2. converting a value to binary
3. binary to integer
4. AND operator
5. OR operator
6. XOR operator
7. manual computation using those operator
8. complement operator and formula
9. right shift operator
10. manually shifted values to right side and got the output
11. formula for right shift operator
12. formula for left shift operator
13. what are special operators
14. how to find memory address of the data type
15. how to compere the memory of the data type
16. membership operator
17. how can you find a value is present it is inerrable or not
18. how to calculate the value if i have more than 2 left shift operator or right shift operators or both